

ECON 351 - Marginal Utility Guide

Total Utility, Marginal Utility, and the Second Derivative

What this guide covers:

Total Utility (TU) - overall satisfaction from wealth or goods

Marginal Utility (MU) - extra satisfaction from one more unit (1st derivative)

Second derivative - whether MU is increasing, decreasing, or flat

Graphs for concave, linear, and convex utility (risk averse / neutral / loving)

1. Total Utility (TU)

Total Utility is the overall satisfaction derived from a given amount of wealth or goods.

- $TU(x)$ = total happiness at quantity x
- Shown as the height of the utility curve

Example: $TU(1) = 10$ and $TU(2) = 25$ means 2 units give 25 total utils.

2. Marginal Utility (MU) - First Derivative

Marginal Utility is the additional satisfaction gained from getting exactly one more unit.

- $MU = u'(x) = dU/dx$
- Discrete: $MU \text{ at unit } 2 = TU(2) - TU(1)$
- MU is the slope of the Total Utility curve

Example: $TU(1)=10, TU(2)=20 \rightarrow MU \text{ at unit } 2 = 10 \text{ additional utils.}$

3. Second Derivative - Is MU Increasing?

The Second Derivative measures the rate at which marginal utility is changing. It tells you whether satisfaction from each extra unit is accelerating, slowing down, or staying flat.

- Second derivative > 0 -> MU is INCREASING
- Second derivative < 0 -> MU is DECREASING
- Second derivative $= 0$ -> MU is FLAT

KEY EXAMPLE (increasing marginal utility):

MU at unit 1 = 10 utility increase

MU at unit 2 = 20 utility increase

MU went from 10 to 20 -> marginal utility is increasing -> $u'' > 0$ -> risk-loving

4. Risk Attitudes from Utility Shape

Match the shape of utility to MU and the sign of the second derivative:

NEGATIVE ($u'' < 0$): Concave utility. MU is going DOWN. Second derivative is negative because MU is decreasing. -> Risk averse

POSITIVE ($u'' > 0$): Convex utility. MU is going UP. MU increases faster every unit. -> Risk loving

NEUTRAL ($u'' = 0$): Linear utility. Flat MU line. Second derivative is 0. -> Risk neutral

5. Graph: Risk Averse (Concave Utility, $u'' < 0$)

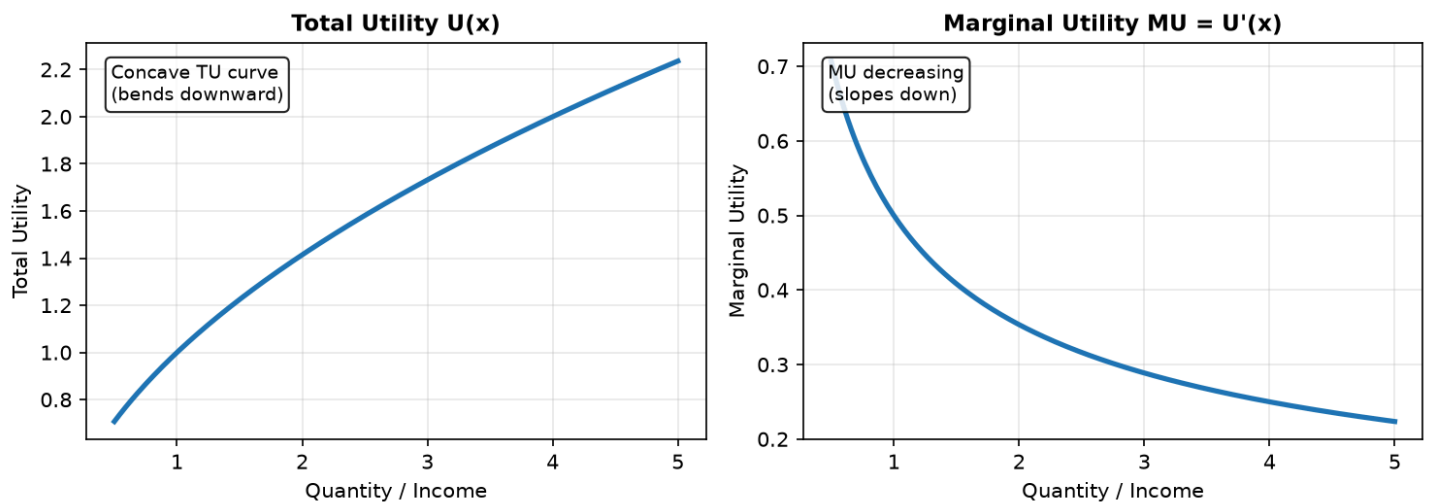
Utility function example: $U(x) = \sqrt{x}$

Total Utility: concave curve (bends downward)

Marginal Utility: decreasing line (each extra unit adds less)

$u'' < 0$ because MU is falling as x rises

Risk Averse: Concave Utility ($u'' < 0$)



6. Graph: Risk Neutral (Linear Utility, $u'' = 0$)

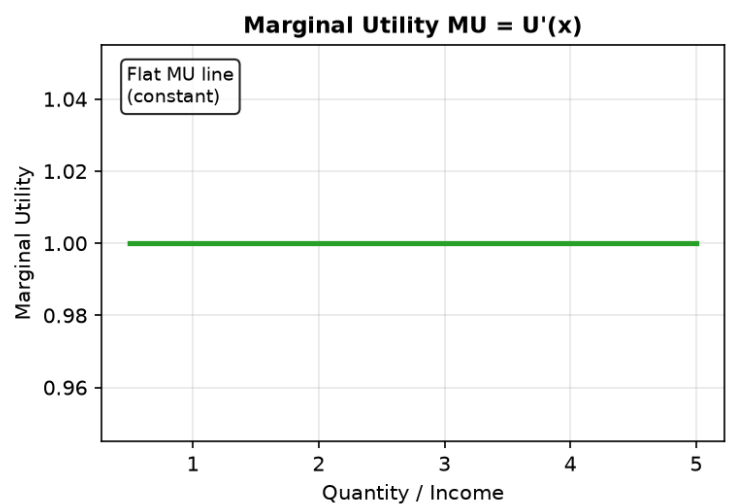
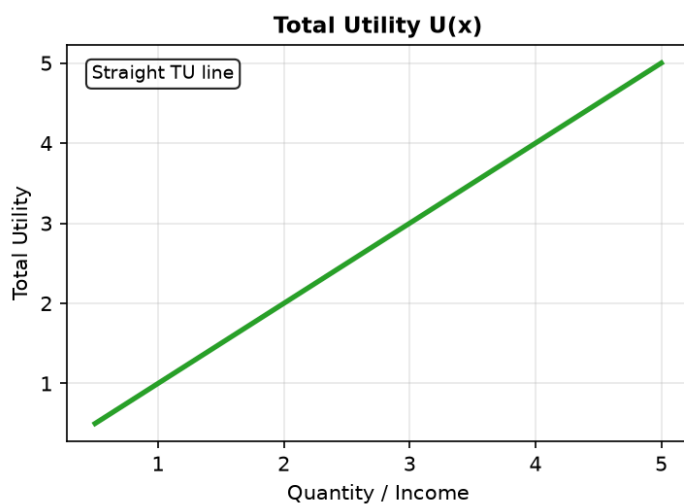
Utility function example: $U(x) = x$

Total Utility: straight line

Marginal Utility: flat horizontal line (constant MU every unit)

$u'' = 0$ because MU never changes

Risk Neutral: Linear Utility ($u'' = 0$)



7. Graph: Risk Loving (Convex Utility, $u'' > 0$)

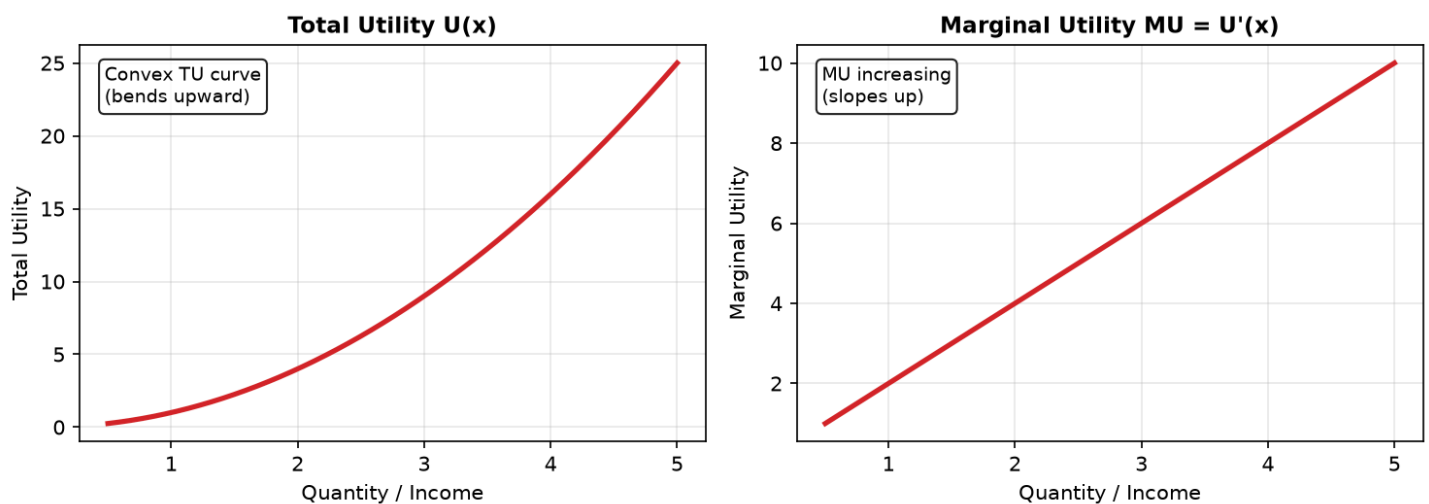
Utility function example: $U(x) = x^2$

Total Utility: convex curve (bends upward)

Marginal Utility: increasing line (each extra unit adds MORE)

$u'' > 0$ because MU is rising - e.g. MU at 1 could be 10, at 2 could be 20

Risk Loving: Convex Utility ($u'' > 0$)



8. Quick Reference Table

Shape	MU behavior	u'' sign	Risk attitude	Example $U(x)$
Concave	MU decreasing	Negative	Risk averse	$\sqrt{x}$
Linear	MU flat	Zero	Risk neutral	x
Convex	MU increasing	Positive	Risk loving	x^2

Remember: second derivative tells you if MU is increasing or decreasing.

If MU at 1 = 10 and MU at 2 = 20, MU is increasing -> convex utility -> risk-loving.